

Google Classroom Code: **qb53eogc**

- Please register by Today (i.e. August 17, 2026) midnight since all communication will be done through GC.
 - IF NOT registered by midnight today, A PENALTY OF 0.5 MARKS will be applied.
 - There are 124 students registered in this course.
 - Project: Build a system using vibe coding
 - Choose any one project, 4 students/project, Make your own group by August 19,
 - No further extension will be granted for evaluation
 - Projects will be evaluated by the TAs on the evaluation day $T + 2$ days.
 - On each evaluation day T , 4-5 groups will be randomly chosen to present/demo their project in the classroom. If the group is absent or not ready, will get a penalty of -5 (minus five) marks.
 - Project-Part A: 15 marks (Traditional Data Integration) [T: Sept 16]
 - Project-Part B: 15 marks (Data Integration under uncertainty) [T: Oct 14]
 - Project-Part C: 10 marks (Data + Content Integration and querying through NLQ) [T: Nov 23]
 - Class quizzes: 10 marks (number of quizzes are not fixed)
 - Class interaction, research paper discussions and assignments: 10 marks
 - Mid-sem exam: 15 marks
 - End-sem exam: 25 marks
- You can reach out to me any day for project discussion. Send me an email first to schedule the online/offline meeting.

Project 1: Identification of uninsured vehicles

- Identification on uninsured vehicle while they are on road
 - Use case: Given a vehicle on the road, we would like to know the complete history (view) of this vehicle [whether this vehicle is insured or not or insurance has expired, (2) when it was registered, etc
- Create multiple structured data sources having information about (1) capturing vehicle registration (car number plate) and other description of vehicle (2) vehicle insurance details (3) vehicle registration details (4) vehicle theft and shredding information (5) report to the Ministry of Transportation if the vehicle is not insured or any suspicion.
 - Design the schema of each DB, populate data, assume you have at least one column with same value (the column/attribute name may be different) since these databases are designed in isolation

Goal: From heterogeneous data to intelligent, trustworthy and decision-centric integration

Project 2: Identification of Asali or Nakali items

- Identification of fake items (medicine) while purchasing
 - Use case: You are purchasing an item, we would like to know the complete history (view) of this item [whether this item is genuine or not (2) when it was procured (3) who was the distributor/supplier
- Create multiple structured data sources having information about (1) capturing item unique code and other description of item (2) distributor/supplier details and items they are distributing to vendors (3) item details by manufacturer (company) with unique code (4) report to the Ministry of Consumer Affairs if the item is Nakali
 - Design the schema of each DB, populate data, assume you have at least one column with same value (the column/attribute name may be different) since these databases are designed in isolation

Goal: From heterogeneous data to intelligent, trustworthy and decision-centric integration

Quiz 1 (one mark)

Think and write any innovative information integration application, as an use case, which has not been discussed in the class. **Particularly address (1) What is the application (what does the application do)? (2) why is it important and what business value does integration create?, and (3) who will be using it for what purpose?**

- Don't propose any design solution

Course vision

- **Traditional view:** combine heterogeneous data sources.
 - Information Integration refers to a category of middleware which lets applications access data as though it were in a single database, whether or not it is.
- **Modern view:** integration is reasoning about meaning, identity, uncertainty and evidence.
 - It enables the integration of data and content sources to provide real-time read and write access, to transform data for business analysis and data interchange, and to data placement for performance, currency and availability.
 - The goal of data integration: tie together different sources, controlled by many people, under a common schema.
- **Future view:** LLMs and agents can participate in integration—but must be verified.
- **End goal:** integrated information that improves decisions, not merely a larger dataset.

THINK DIFFERENTLY

When two sources disagree, how should an intelligent system decide what to believe?

Clue: The answer will evolve throughout the course.

Discussion pattern for each topic

- We will start with a real-world case study.
- We will take one design question/problem (start with a contradiction or failure).
- You need to propose the solution(s) 'Think Differently', justifying your algorithm/ explanations.
- Think for a business-value argument alongside the technical solution.
- We will introduce the formal model / algorithm / architecture.
- You need to revisit the original question and challenge the assumptions.
- End with a research question: what remains unsolved?

Information Integration Approaches and Architectures

W2 reading: Automatic End-to-End Data Integration using Large Language Models

Acknowledgement: This lecture includes contents from many open sources.

The central philosophy of IIA

- Integration is not merely joining data.
- Integration is not merely resolving schemas.
- Integration is not merely matching entities.
- Integration is the process of constructing defensible knowledge from heterogeneous evidence.
- The next generation of integration systems will be semantic, probabilistic, graph-based, multimodal, LLM-assisted, privacy-aware and decision-centric.

THINK DIFFERENTLY

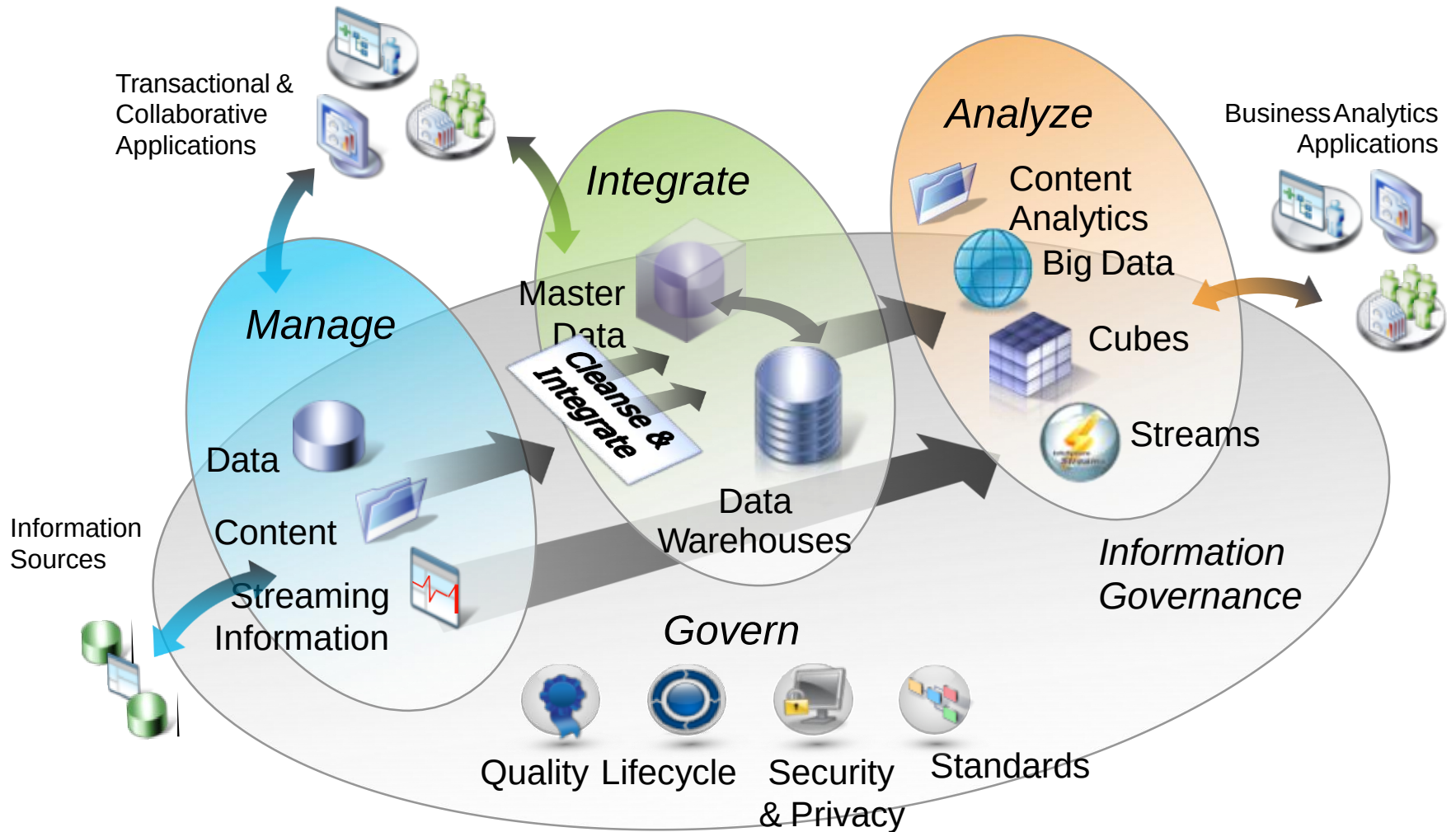
The final challenge: What should an intelligent integration system do when it does not know the answer?

Long-Standing Challenges in Information Integration

- Numerous works in the past 30 years
- In many communities: DB, AI, KDD, Web, Semantic Web



Mastering information across the Information Supply Chain

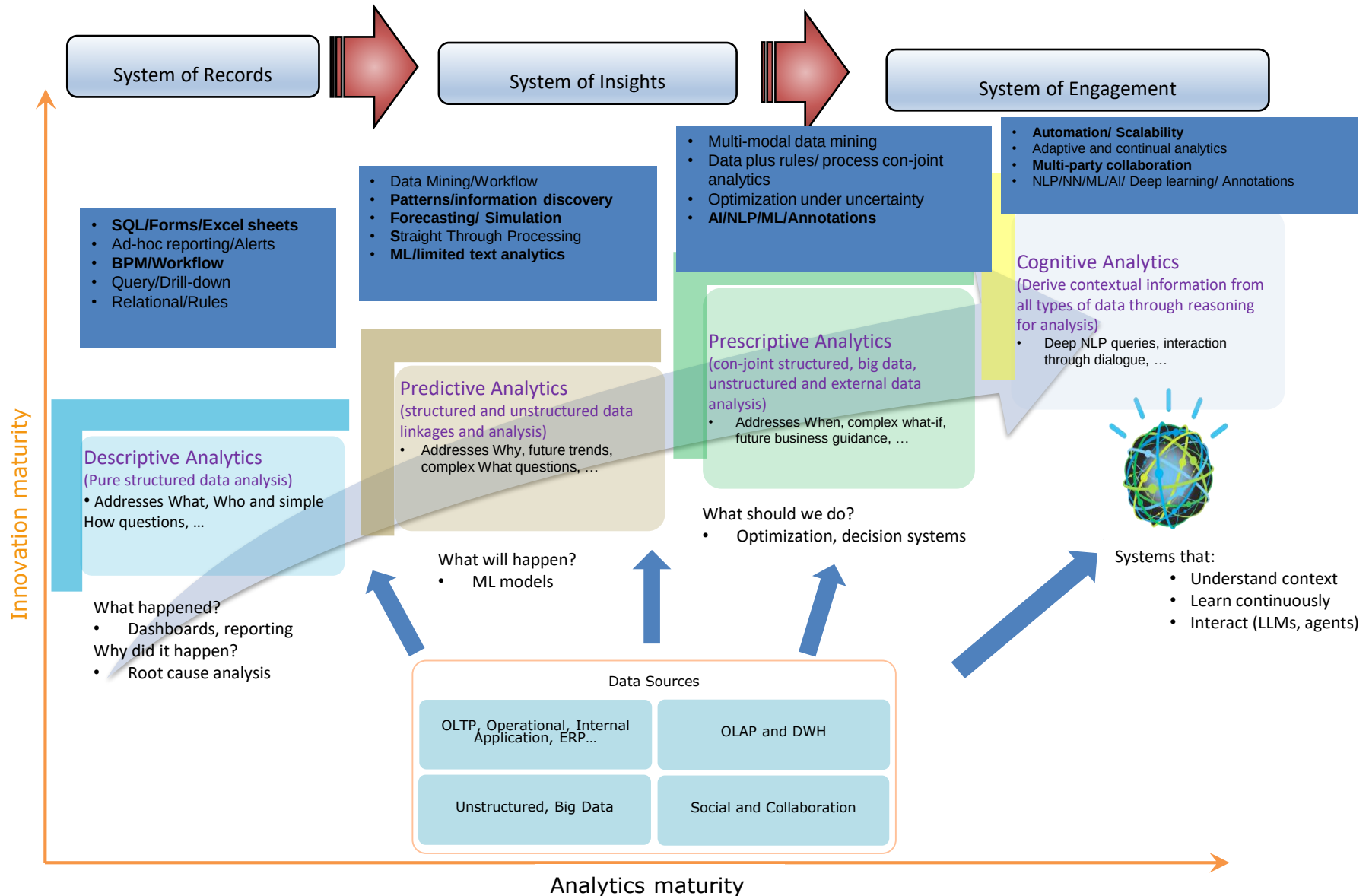


Trusted ♦ Relevant ♦ Governed

A unifying architecture

- **Sources**: databases • APIs • documents • knowledge graphs • sensors • images.
- **Discovery & semantics**: **profiling** • schema matching • ontology alignment • embeddings.
- **Identity**: entity resolution • graph evidence • temporal identity.
- **Evidence**: provenance • quality • reliability • uncertainty • truth discovery.
- **Intelligence**: LLMs • RAG • agents • human feedback.
- **Outcome**: trustworthy integrated information → decisions.

Data Querying and Analytics ... then and now, a continuous journey: AI & Data Science Perspective



Integration Approaches

- Materialized-based
 - Extract → Transform → Load into DWH
 - Best for governed BI, historical analytics and repeatable reporting
- Mediation-based: Virtual integration via federated queries
 - Sources → Wrappers → Mediator → Federated Query
 - Best for real-time access and autonomous source systems
- Hybrid: Semantic layer + materialized views
 - Useful for enterprise semantic architectures
- Straight Through Processing: Rule-based/Learning-based
 - Event → Rules/ML → Decision → Automated Action
- AI/ML-driven: Use embeddings for schema & entity mapping
 - Embeddings → candidate generation → ML/LLM decision → validation
- Context-Oriented: Entity Discovery, enrichment and linkages
 - Entity Discovery → Context collection → Enrichment → Linkage → Contextual view

How to compare integration approaches

Four design questions

- Where does integration happen?
- When does data become materialized?
- Where is semantic/identity reasoning performed?
- What happens when sources change or fail?

Central trade-off

- Latency vs freshness
- Coupling vs autonomy
- Control vs flexibility
- Engineering cost vs integration intelligence

No architecture is universally superior: the application determines the right integration strategy.

The recurring business-value framework

- Accuracy — Did we get the right answer?
- Completeness — Did we integrate the necessary evidence?
- Trust — Can we explain and defend the answer?
- Cost — Is the integration economically worthwhile?
- Decision value — Did integration improve an outcome?

THINK DIFFERENTLY

Which of these five dimensions should dominate when they conflict?

Clue: There is no universal answer; the application determines the utility function.

1. ETL / ELT — materialized integration

Sources

ERP • CRM
• CSV • APIs

Extract

Batch / CDC

Transform

Clean • map
• join

Load

DWH

Serve

BI • analytics
• ML

Best-suited IIA applications

- Enterprise BI and reporting
- Regulatory / historical reporting
- Stable analytical workloads
- Cross-source aggregates

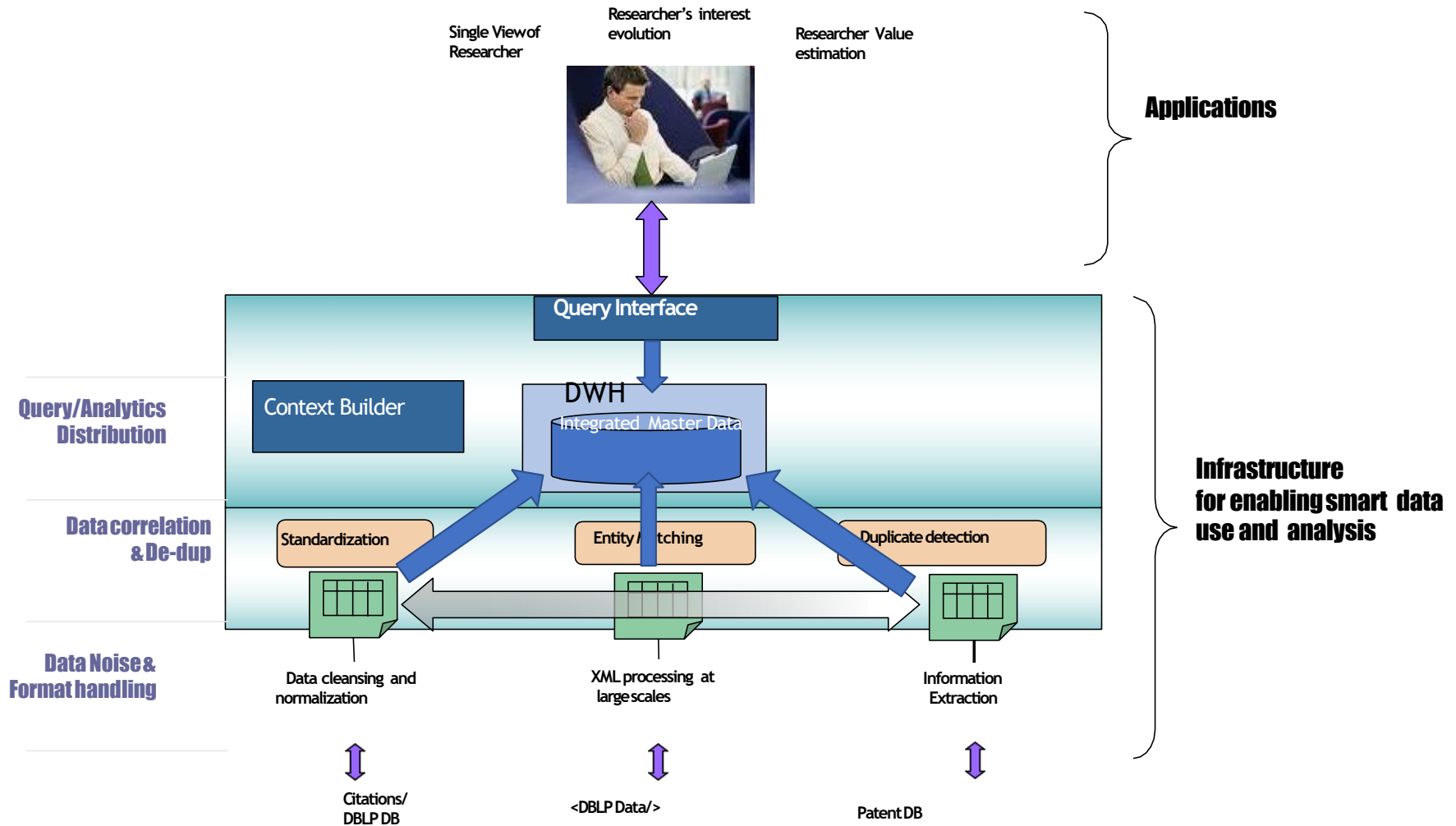
Key trade-offs

- Pipeline latency
- Brittle transformation logic
- Schema changes propagate
- Storage duplication

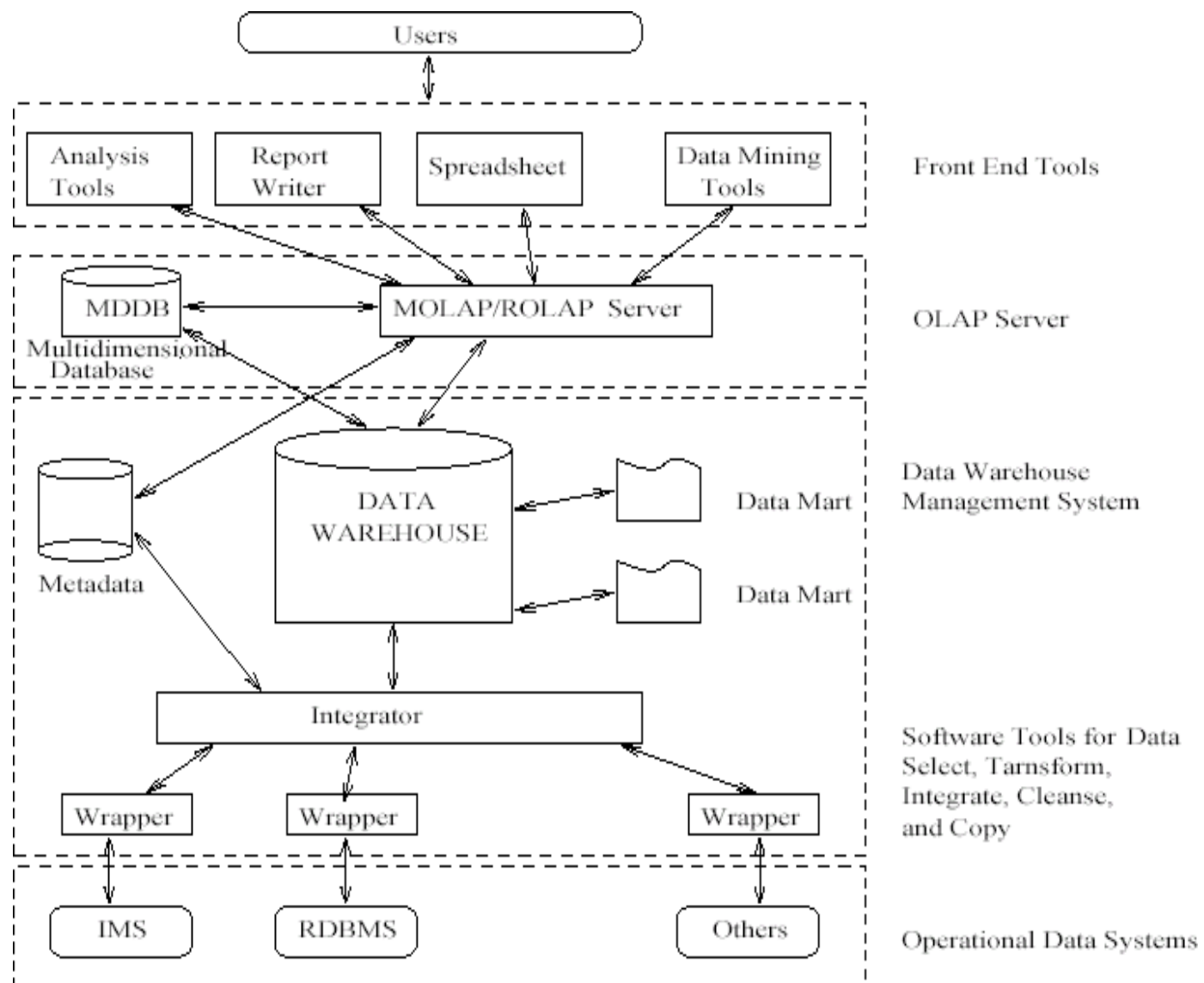
Design principle

- Materialize integrated data when repeatable analytics and governed history matter most.

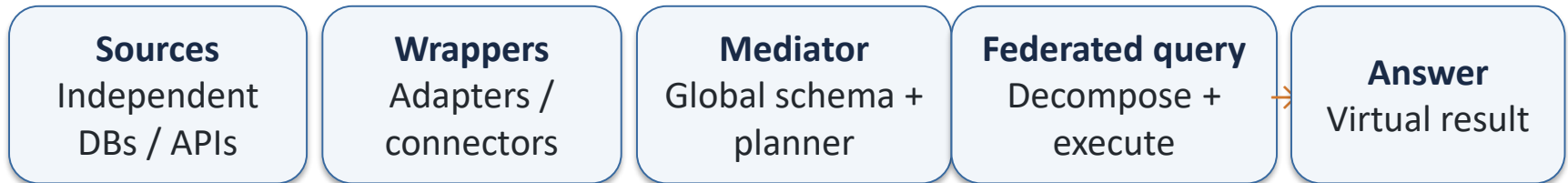
Materialization Approach



II Architecture: A Data Warehousing Approach



2. Mediation / Federated Virtual Integration



Best-suited IIA applications

- Real-time operational views
- Autonomous systems
- Avoiding data copies
- Ad-hoc cross-system queries

Key trade-offs

- Runtime latency
- Source availability dependency
- Complex query planning
- Cross-source consistency

Design principle

- Keep data at the source; integrate at query time.

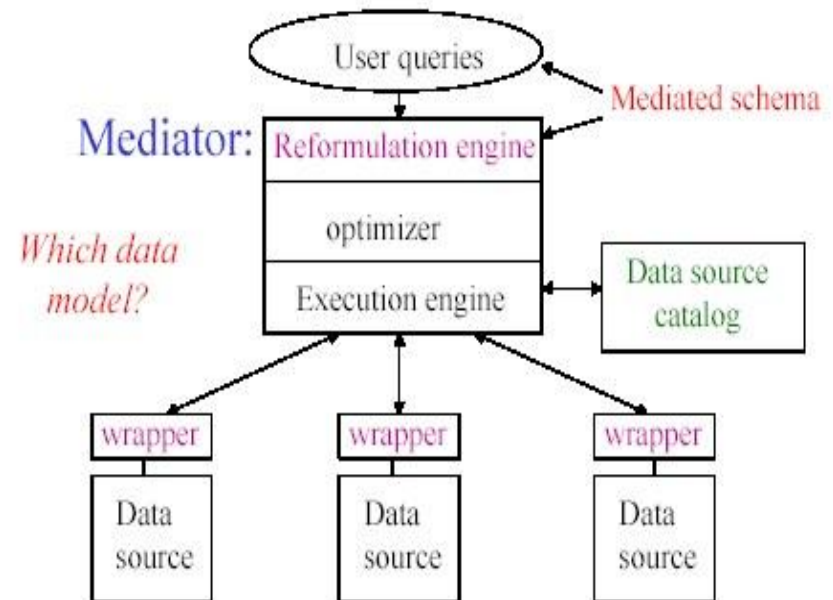
Virtual integration favours freshness and source autonomy over predictable performance.

II Architecture: Virtualization Layer Approach

Workflow:

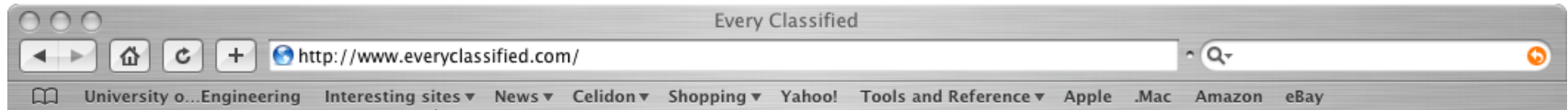
Query → decomposition → wrapper invocation → integration

- Leave the data in the sources.
- When a query comes in:
 - Determine the relevant sources to the query
 - Break down the query into sub-queries for the sources.
 - Get the answers from the sources, and combine them appropriately.
- Data is fresh. Approach scalable
- Issues:
 - Relating Sources & Mediator
 - Reformulating the query
 - Efficient planning & execution



Garlic [IBM], Hermes[UMD];Tsimmis,
InfoMaster[Stanford]; DISCO[INRIA];
Information Manifold [AT&T];
SIMS/Ariadne[USC];Emerac/Havasu[ASU]

Integration for creating a single site to search for jobs/rentals/...



 **EveryClassified** beta

About EveryClassified Search

AUTOS

 Research new and used makes and models to find the right car for you.

Make:

Model:

Year:

Location:
(Enter City, State or Zip Code)

RENTALS

 Search for apartments or houses for rent.

Type:

From: to

Bed:

Location:
(Enter City, State or Zip Code)

REAL ESTATE

 Make finding your next home easier. Search homes for sale.

From: to

Bed: Bath:

Location:
(Enter City, State or Zip Code)

JOBS

 Find a job.

Keywords:
(ie. Retail, Sales, Management)

Location:
(Enter City, State or Zip Code)

PERSONALS

 Find that special someone.

I am a

seeking a

Age to

Location:
(Enter City, State or Zip Code)

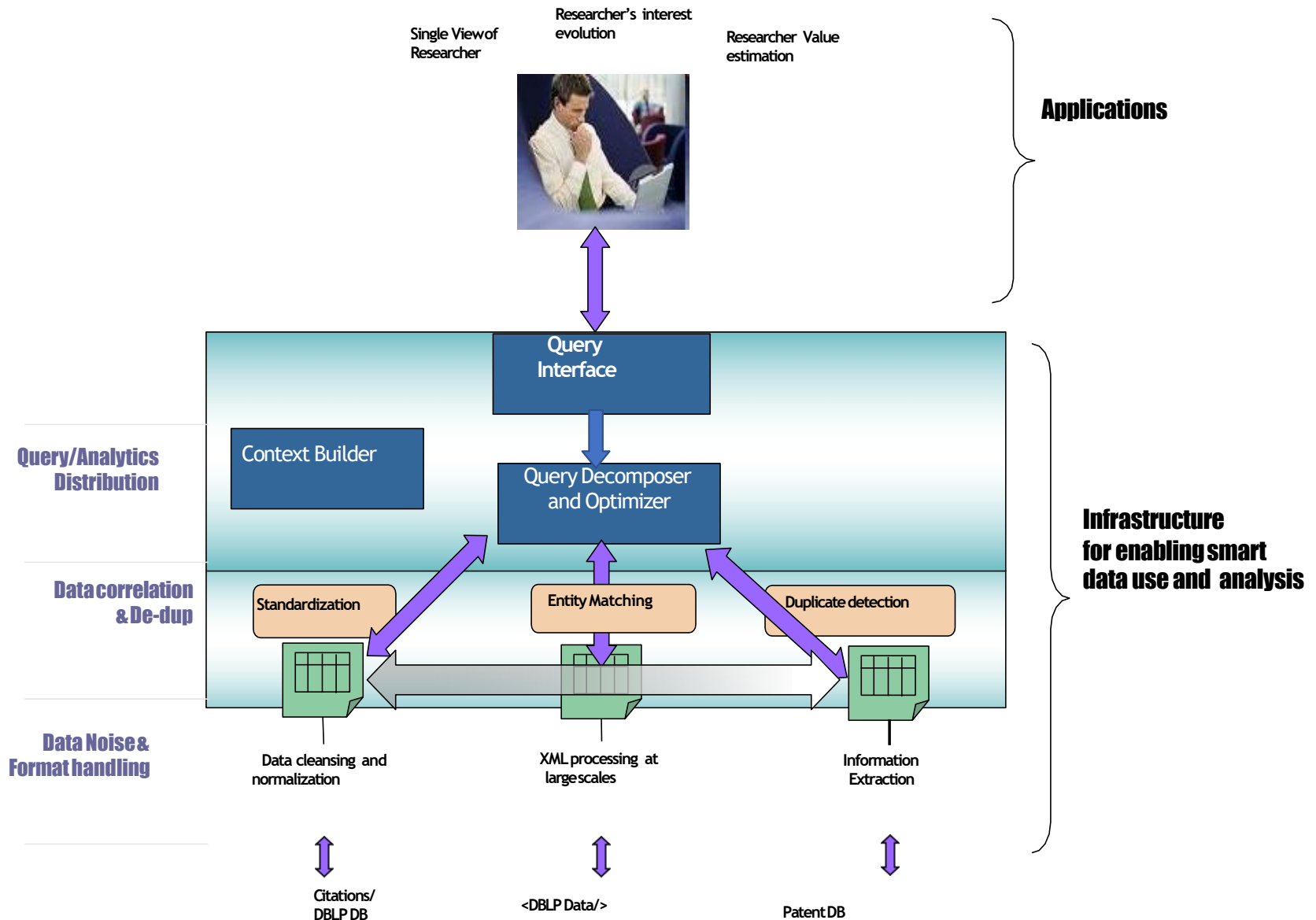
GENERAL

 Search for miscellaneous items for sale.

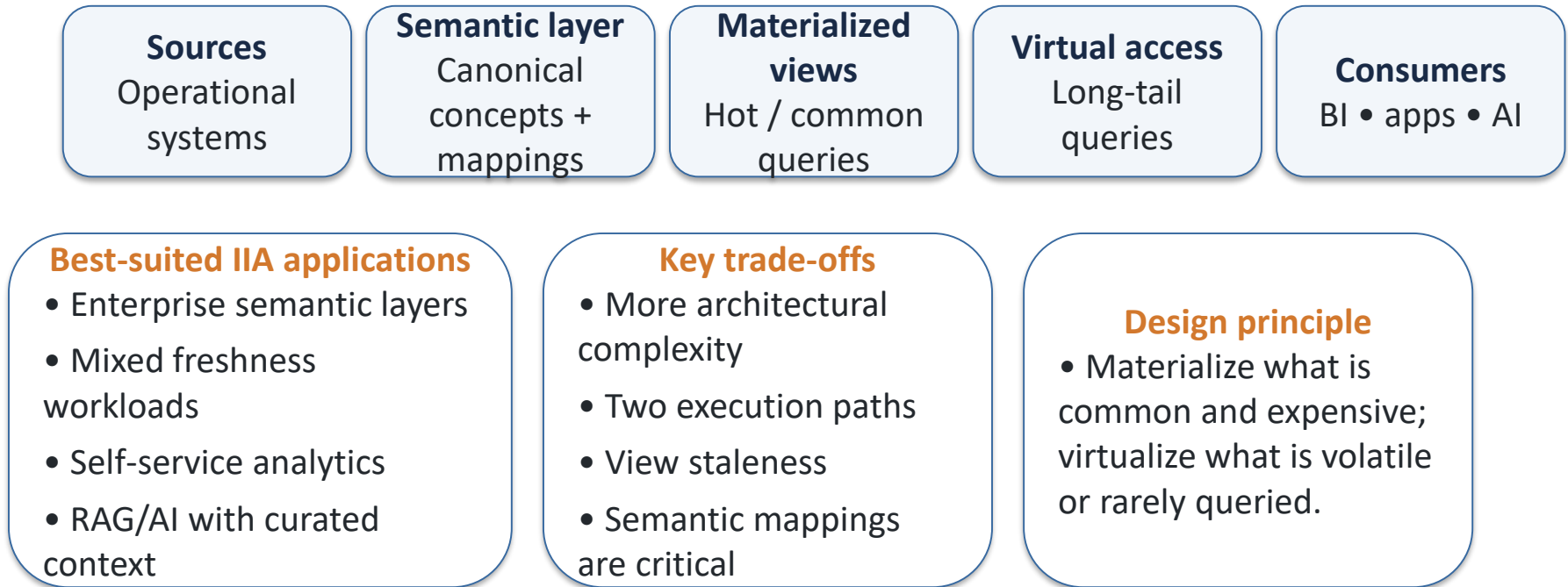
Keywords:
(ie. furniture, Wood, Bookshelf)

Location:
(Enter City, State or Zip Code)

Integration through (sub-)query federation (Mediator Approach)



3. Hybrid — Semantic Layer + Materialized Views



Hybrid is often the practical enterprise compromise between ETL and federation.

4. Straight-Through Processing (STP)

Event / record

Transaction •
message •
stream

Rules

Deterministic
mapping /
validation

Decision

Route • enrich
• accept

Learning

Optional
classifier /
model

Action

Process
automatically

Best-suited IIA applications

- Real-time operational integration
- Order-to-cash workflows
- Fraud / eligibility pre-screening
- Event processing

Key trade-offs

- Rule explosion
- False positives/negatives
- Model drift
- Human exception handling

Design principle

- Automate the common path; route uncertainty to human review or richer reasoning.

STP is a processing strategy: rules, ML, or both can implement it.

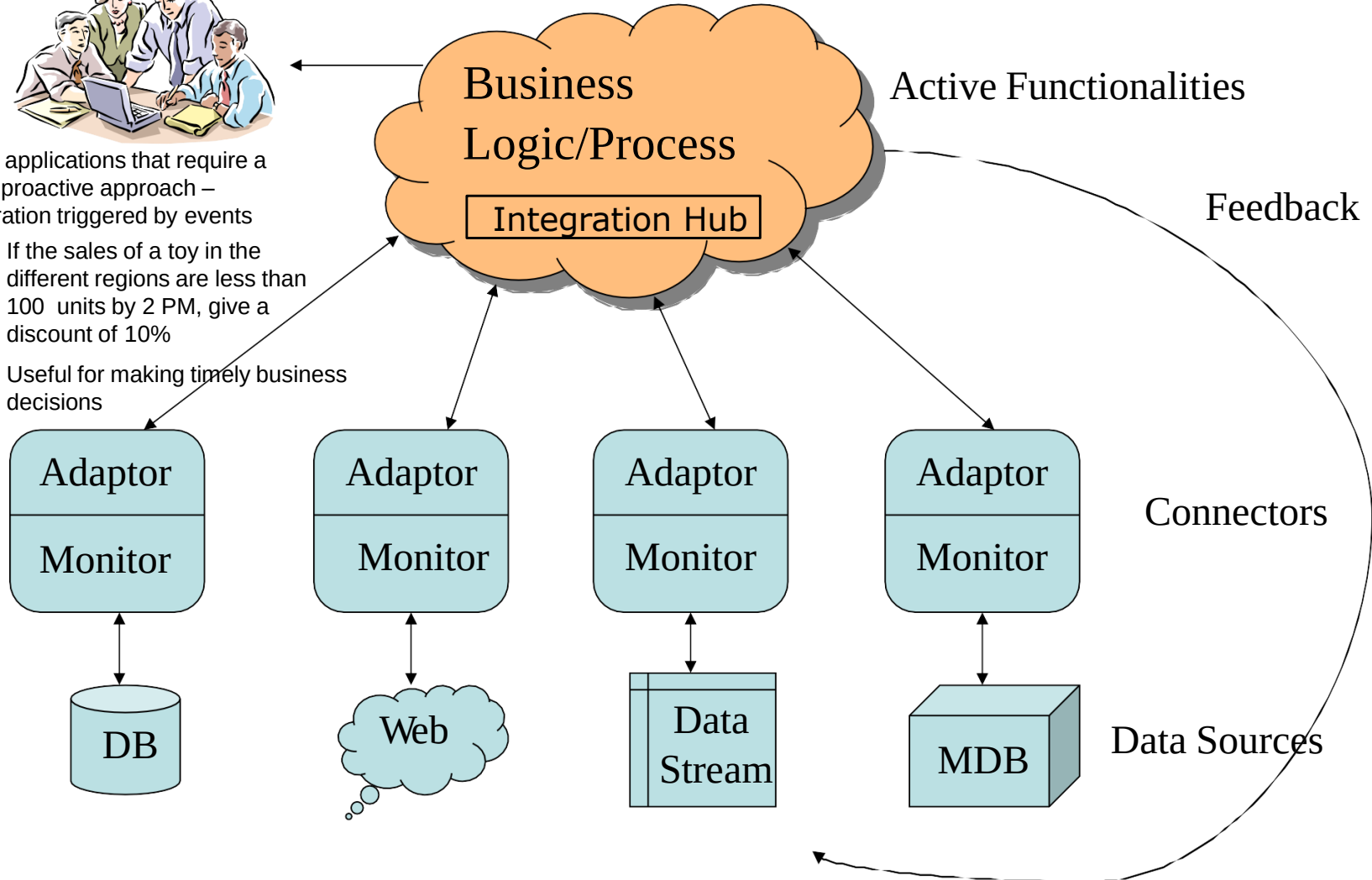
Event based Information Integration for STP

Purpose: Real-time integration without manual steps

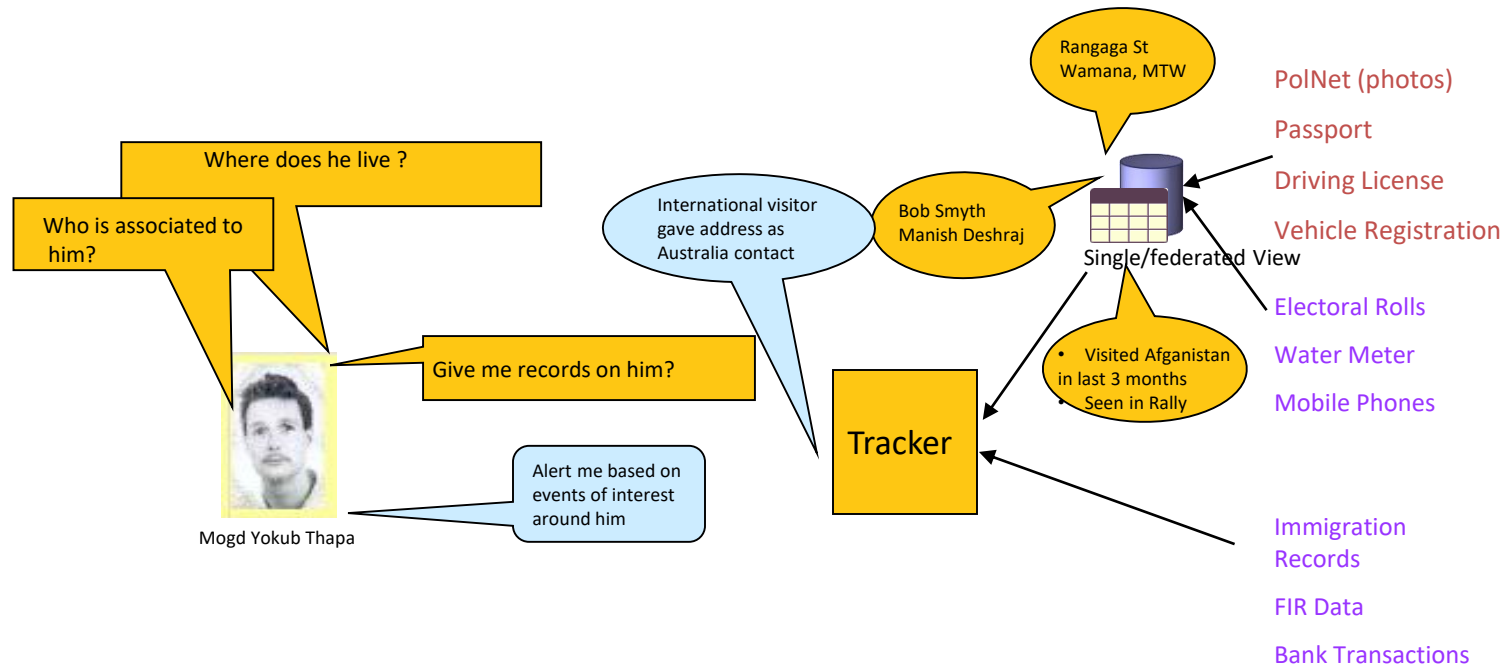


- Many applications that require a more proactive approach – Integration triggered by events

- If the sales of a toy in the different regions are less than 100 units by 2 PM, give a discount of 10%
- Useful for making timely business decisions



Data Finds Data: Entity relationship discovery



Challenges

- Data collection and maintenance (with data quality)
- Information extraction
- Multi-modal data integration
- Entity Matching (with privacy preserving)
- Integrated Analysis and Intelligence

5. AI/ML-driven Integration — embeddings + models

Sources

Schemas •
records • text

Representation

Embeddings /
features

Candidate generation

Semantic
similarity

Decision

ML / LLM +
evidence

Outcome

Accept •
review • learn

Best-suited IIA applications

- Schema matching
- Entity resolution
- Document/multimodal integration
- Poorly documented data

Key trade-offs

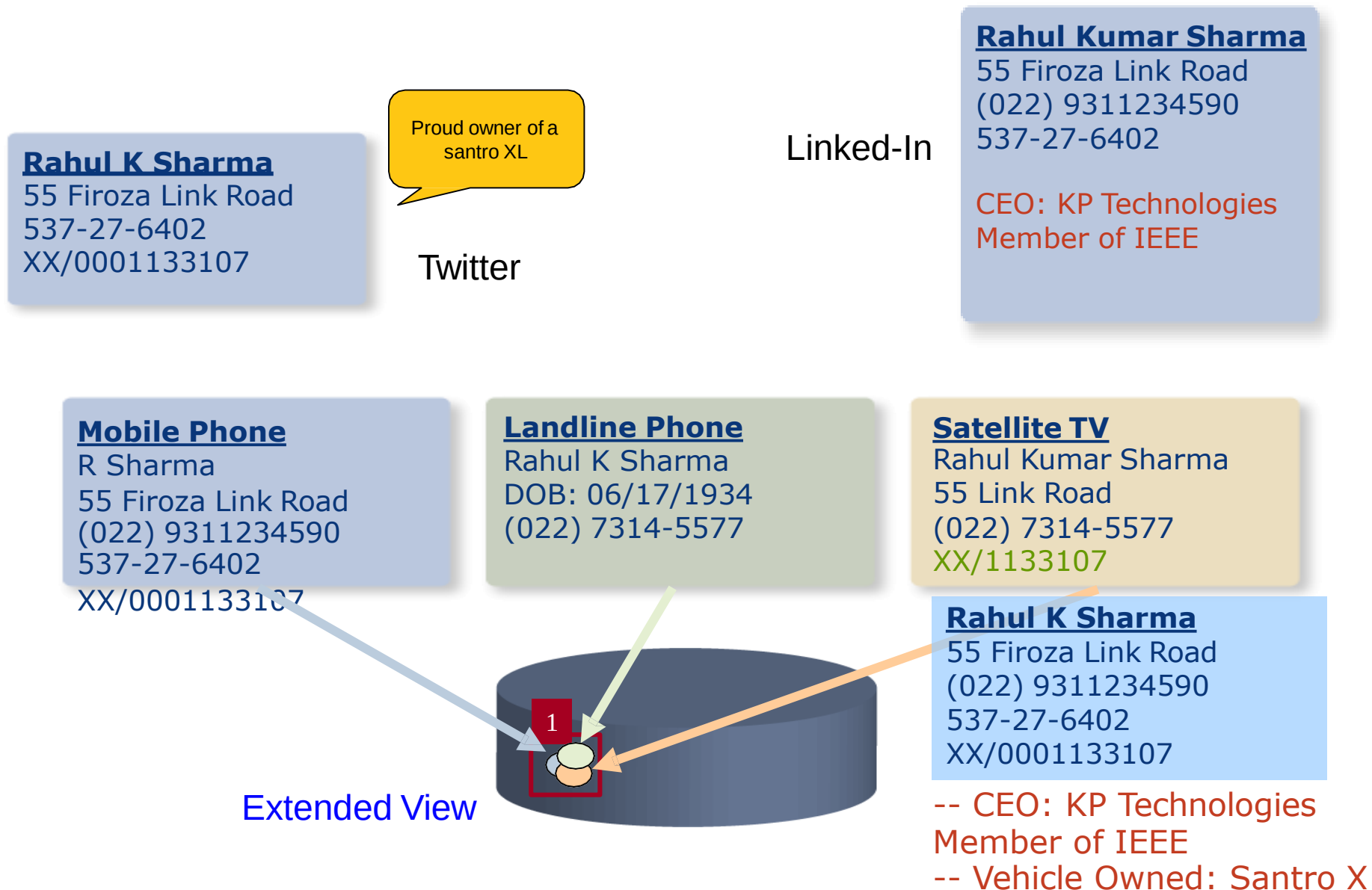
- Similarity $\neq$ equivalence
- Bias and drift
- Explainability challenges
- False matches propagate

Design principle

- Use embeddings to generate candidates; use evidence, constraints and validation to make decisions.

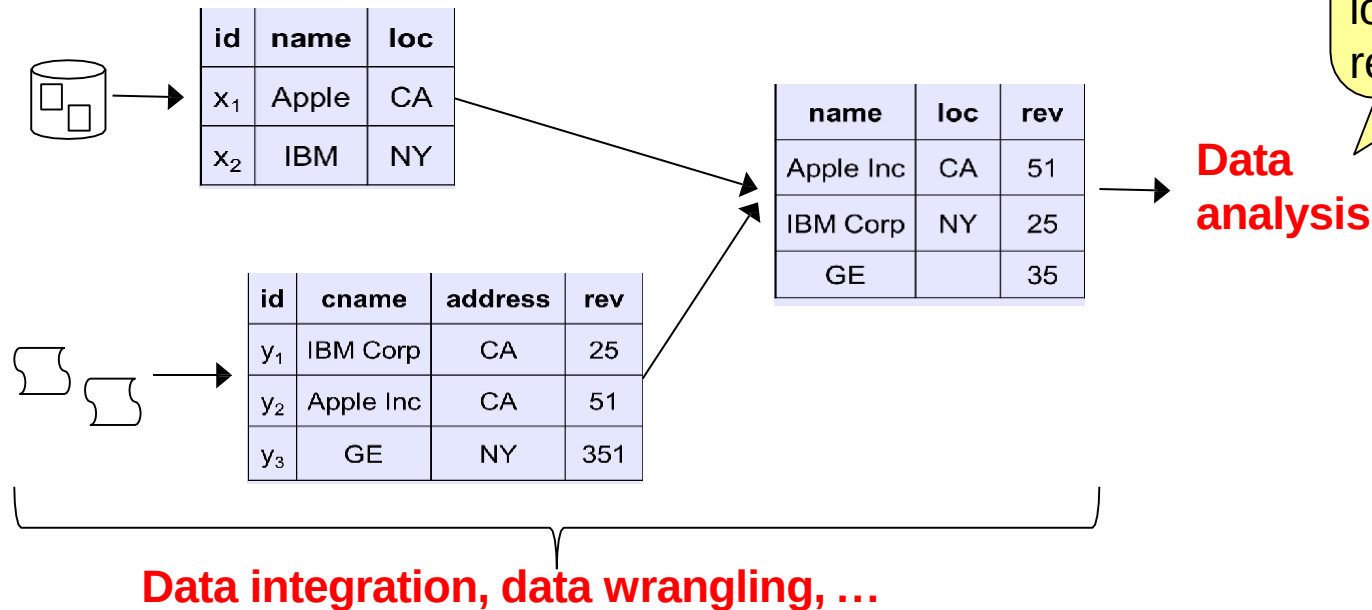
AI is strongest where semantic variation is high and deterministic rules alone are insufficient.

Integration for Single truth --

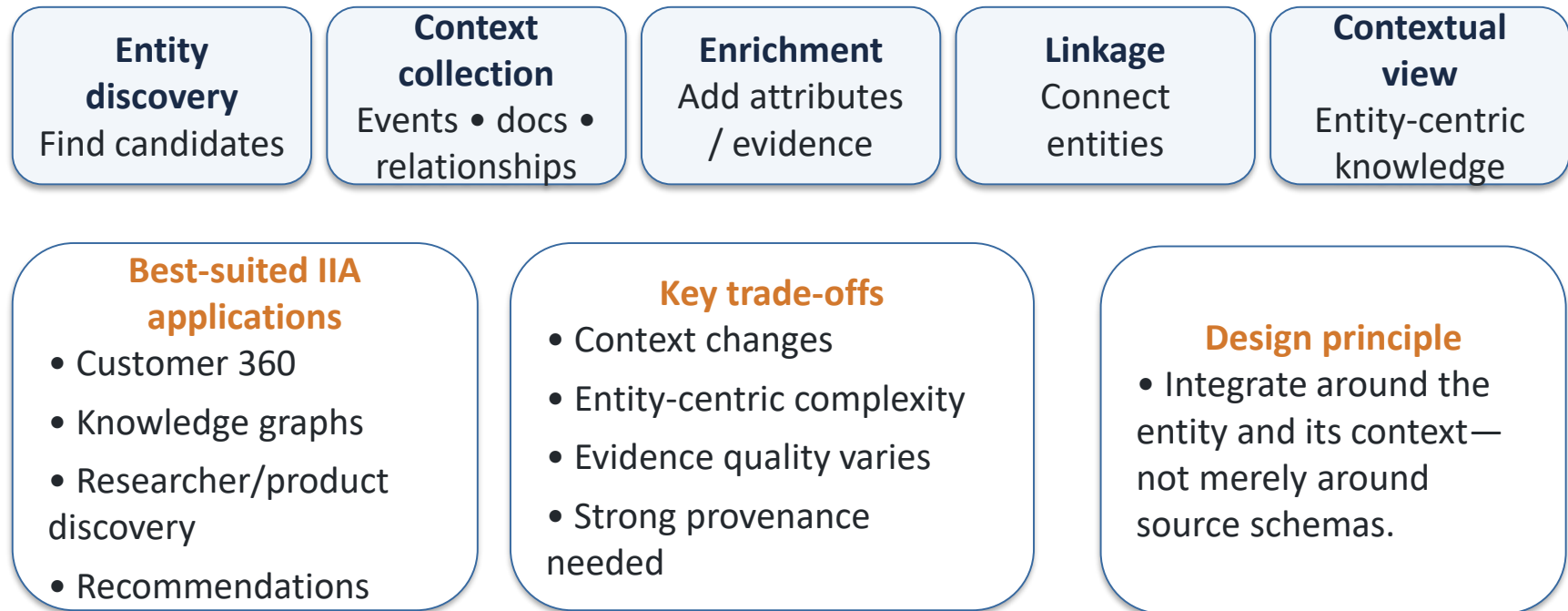


'Text + Data' Integration

- The raw data to insight pipeline



6. Context-Oriented Integration



Context-oriented integration is a natural architecture for KG-style applications.

Information Integration in Google search --

Inbox (1,244) - anhaid@g

apple - Google Search

https://www.google.com/search?q=apple&oq=apple&aqs=chrome..69i57j0l5.1228j0j4&sourceid=chrome&ie=UTF-8

Apps M WSJ bogle myself intra public dsuwmadison BigGorillaInternal BigGorilla my-hp Other bookmarks

Google apple

All News Maps Images Shopping More Search tools

About 1,850,000,000 results (0.80 seconds)

Official Apple Site

Ad www.apple.com/

Shop iPhone, iPad, Mac, **Apple TV & Apple Watch**. Learn more & shop n...

Apple Back to School ... iPhone 6s MacBook iPad Pro

Apple

www.apple.com/ Apple

Apple leads the world in innovation with iPhone, iPad, Mac, **Apple Watch**, iOS, OS X, watchOS and more. Visit the site to learn, buy, and get support.

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Mac

MacBook Pro - MacBook - MacBook Air - iMac - Mac

iPhone

Explore the world of iPhone. Check out iPhone 6s, iPhone 6 ...

iPad

Explore the world of iPad. Check out iPad Pro, now in

Apple

Technology company

apple.com

Apple Inc. is an American multinational technology company headquartered in Cupertino, California, that designs, develops, and sells consumer electronics, computer software, and online services. Wikipedia

Stock price: AAPL (NASDAQ) \$106.94 -0.63 (-0.59%) Aug 26, 4:00 PM EDT - Disclaimer

Founded: April 1, 1976, Cupertino, CA

Headquarters: Cupertino, CA

Sales: 1 (800) 692-7753

Products: iPhone, iPad, iPod, Macintosh,

IMG_2358.JPG

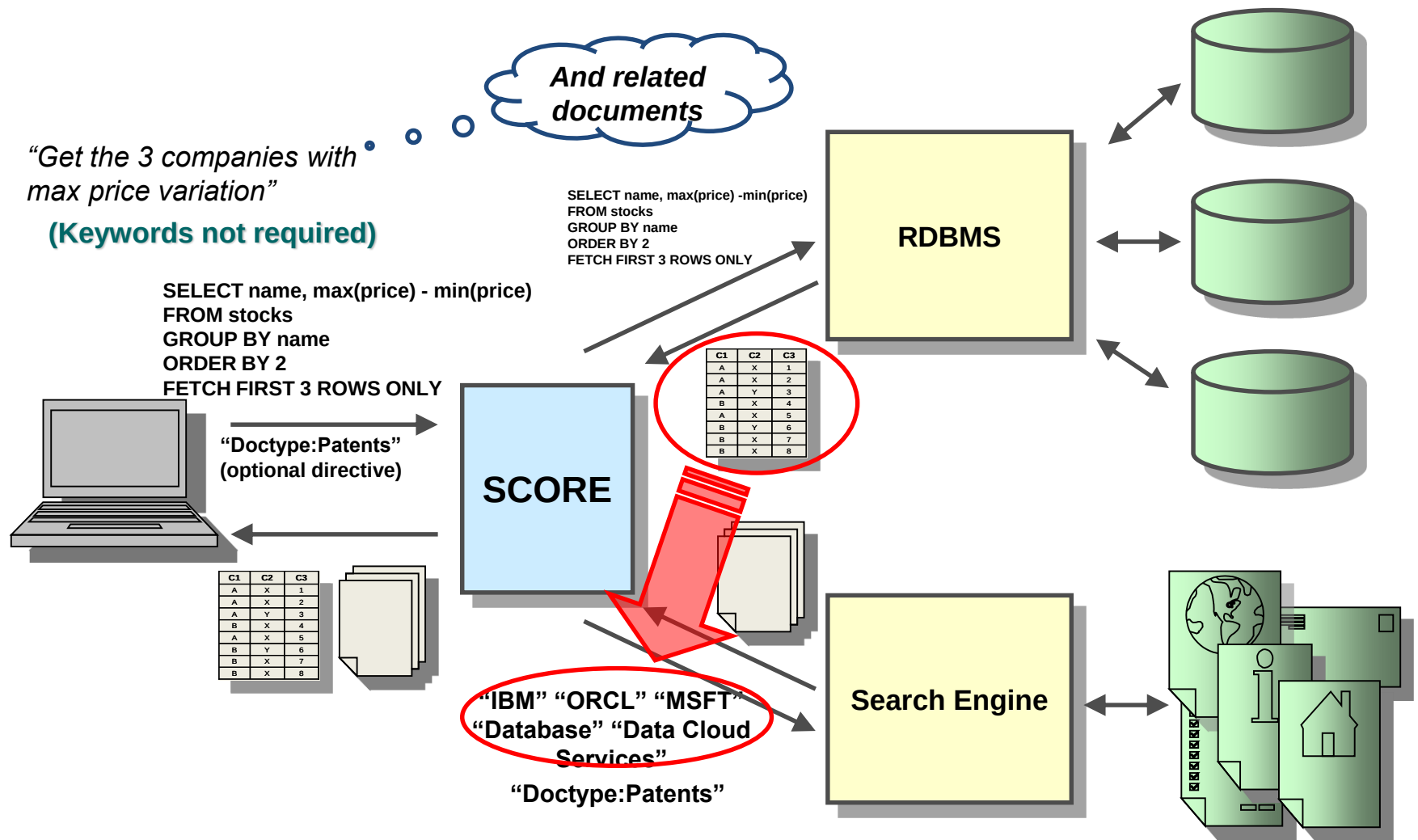
magelan-vldb16-v5.pptx

Show all downloads...

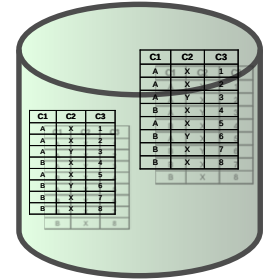
I'm Cortana. Ask me anything.

4:52 PM 8/26/2016

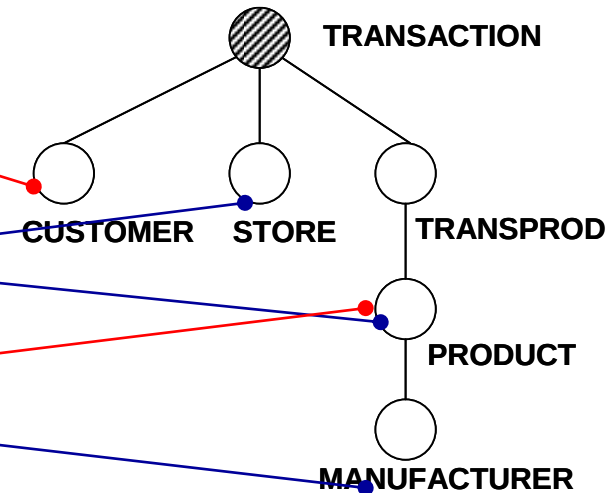
Integration of Structured Query Results with Unstructured Data



Integrating Unstructured Documents with Structured Data



I am <Name> Bharat Kumar </Name>
....
..... bought a
<Company>Sony</Company>
<product> DVD player </product>
....
from <Company>JK Electronics
</Company>



Additional “sidebar”
information available
as a result of the
annotation

CustId	StoreId	Payment	Discount	Terms
A756K9	S8976	Card (AMEX)	Promo# 1236	NOREFND

CustId	Name	Loyalty	Club	Addr
A756K9	Bharat Kumar	Platinum	Royal	Okhla Phase 3, New Delhi

Integrating Real-time Audio with Databases

Call Assist

Call Progress 134 Seconds

Transcripts

128970.0,129195.0, WITH
129195.0,129450.0, THAT
129450.0,129630.0, NAME
129630.0,130830.0, <silence>
130830.0,131025.0, NOW
131025.0,131355.0, <silence>
131355.0,131670.0, THANK

Noun Phrases

IMPALA
MID
SIZE
PONTIAC
GRAND
UNLIMITED
UNLIMITED
STANFORD

Candidate Tuples

Tuple Rank	model	location	promotion	type	rate	description
1	pontiac grand	Stamford Connecticut	NA	mid size	20	4 door 2 adult
2	pontiac grand	Stamford Connecticut	AAA	mid size	25	4 door 2 adult
3	pontiac grand	Dallas Fort Worth	Club	mid size	20	4 door 2 adult
4	Chevy Aveo or Similar	Dallas Fort Worth	AAA. COSTCO	mid size	18	4 door, 4 people, 2 bags, unlimited mileage
5	Chevy Aveo or Similar	dallas Love field	AAA. COSTCO	mid size	18	4 door, 4 people, 2 bags, unlimited mileage

Suggested Questions

What promotion you prefer? Expected Gain=2.32
What rate you prefer? Expected Gain=2.32
What promotion you prefer? Expected Gain=1.92

Suggested Promotions

Promotion From	To	Maximum Discount(%)	Promo Type
pontiac grand	Chevy Aveo	10.0	Upsell
pontiac grand	Chevy Impala	30.0	Upsell
pontiac grand	Kia sportage	30.0	Upsell
mid size	standard	20.0	Upsell
mid size	minivan	10.0	Upsell
mid size	fullsize SUV	40.0	Upsell

Start

Architecture selection matrix

Best fit

- ETL/DWH → governed BI, historical analytics
- Mediation → real-time federation
- Hybrid → semantic layer + mixed workloads
- STP → operational workflows
- AI/ML → ambiguous schemas/entities
- Context-oriented → discovery, KG, 360° intelligence

Selection criteria

- Freshness
- Query complexity
- Source autonomy
- Volume / latency
- Semantic ambiguity
- Provenance needs
- Tolerance for duplicated storage

Choose architecture based on application requirements, not technology fashion.

Classroom exercise — choose the architecture for your application

Scenario

- Hospital needs a current patient view
- Autonomous clinical systems
- 7 years of historical analytics
- Some identities are ambiguous
- Doctors need live medication status

Ask students

- Which parts use ETL?
- Which parts use federation?
- Where is entity resolution?
- Would a semantic layer help?
- Where is AI appropriate?
- What should be materialized?

Do not accept “one architecture for everything.” Require justification for every boundary.

Think and then answer w.r.t. your application

Think differently

- If an embedding says two entities are 97% similar, should they be merged?
- If a virtual query is always slow, when should it become materialized?
- If the semantic layer is wrong, can downstream applications be correct?
- Can context itself become the integration key?

Research directions

- Adaptive integration architectures
- LLM-assisted schema/ontology evolution
- Learning-based query federation
- Context-aware entity resolution
- Self-healing integration
- Provenance-aware AI integration

Ultimate question: Can an integration architecture dynamically decide how and where to integrate each piece of information?

Key Challenges

1. Understand the data — schemas, semantics, ontologies, context, curation.
2. Schema and Data heterogeneity, and Automated data transformations — schema mapping, data mapping, information discovery.
 - Uniform (or source specific) query access to data sources
 - Distributed query processing and optimization
 - Consolidating, transforming, and mining data for analysis.
3. Determine identity and truth — entity resolution, record linkage, graph evidence, uncertainty, provenance, fusion.
4. Build intelligent integration — graphs, multimodal data, LLMs, RAG, agents.
 - Reduce the effort needed to set up an data curation, integration tasks.
 - Enable the system to perform gracefully with uncertainty (e.g., on the web/noisy source/...)
5. Make integration useful — active learning, fairness, self-healing, decision intelligence.